

Machine Learning Approach for Optimal Football XI Selection Based on Opponent Tactical Analysis

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Abstract

The choice of eleven is one of the most important pre-match decisions in football and there is limited research in machine learning in football lineup prediction in comparison with the vast literature in match outcome prediction. Most of the previous studies only consider player form, opponent strength or formation compatibility separately. We propose an end to end pipeline to co model these aspects, leveraging 25,187 player-match observations from 23 La Liga teams from the 2024-25 and 2025-26 seasons from FBref. After a leakage audit and correlation pruning, 126 of 178 engineered features are retained based on a rolling form, fatigue, head to head history, formation fit and team context. Five classifiers (Decision Tree, MLP, Random Forest, XGBoost and a Stacking Ensemble) are tested using a strict time based protocol with SMOTE applied only to the training fold. XGBoost is the best model with respect to all metrics (validation AUC 0.8654; TimeSeriesSplit AUC 0.8484 ± 0.0126 ; Brier score 0.1304) and has the best probability calibration of all the models. A secondary win probability model (AUC 0.6818) is embedded in a deployment pipeline that looks for eight common formations and suggests an ideal starting XI and match win probability for any upcoming La Liga match-up.

CCS Concepts

• **Computing methodologies** → **Supervised learning by classification**; *Supervised learning by regression*; **Machine learning**; • **Applied computing**;

Keywords

football analytics, starting XI prediction, XGBoost, SMOTE, TimeSeriesSplit, SHAP, La Liga, imbalanced classification

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1 Introduction

Football is the most data intensive team sport in the world, and the global sports analytics market reached USD 5.7 billion in 2025, with projected growth exceeding 18% per year through 2033 [1]. Top European clubs such as Liverpool, Manchester City, Brentford, and Brighton now maintain dedicated data science units whose outputs influence recruitment, in-game tactics, and lineup decisions [2].

Despite this commercial maturity, academic machine learning research in football remains heavily skewed toward two problems: match outcome prediction [3–5] and player valuation [6]. The fundamental pre match decision of selecting the starting eleven has received limited attention. A systematic review of 172 studies published between 2019 and 2024 found that 72% focus on outcome prediction, while only one study addressed lineup prediction, and it excluded opponent context [7].

The lineup decision is the most important tactical choice a manager makes before kickoff. It depends on a complex interaction of player form, fatigue, opponent identity and tactical setup, formation compatibility, and head to head history. Existing approaches typically consider these factors in isolation. Abidin [8] uses Random Forest on coach assessments from a single youth academy; Cortez et al. [9] predict lineups based on physiological variables; Pappalardo et al.'s PlayeRank [10] produces a global player ranking rather than a constrained eleven player lineup; and TacticAI [11] applies graph neural networks to tracking data but is limited to corner kick situations and relies on proprietary datasets.

This paper addresses this gap by developing an end to end pipeline that predicts both the starting eleven and the match win probability using publicly available match event data. The model jointly incorporates (i) position-weighted, exponentially decayed rolling player form, (ii) opponent specific head to head history, (iii) formation compatibility, and (iv) team and opponent rolling context.

We evaluate five classifiers using a time aware validation framework, with SMOTE applied only to training folds to prevent data leakage [12]. Two research questions guide the study:

RQ1: Can a supervised model jointly leverage player performance and opponent tactics to predict the starting eleven?

RQ2: Do opponent and formation features improve lineup prediction compared to models trained only on player statistics?

Both questions are answered affirmatively. The main contributions are: a position weighted rating that emerges as a strong predictive feature; a feature space of 178 leakage free variables across five groups; a rigorous time series evaluation in which XGBoost achieves an AUC of 0.8654 (0.8484 ± 0.0126 across folds); and a deployment pipeline that evaluates eight common formations to recommend an optimal lineup and win probability for any La Liga fixture.

2 Related Work

Outcome prediction is the most prevalent problem type in the football machine learning literature. Hubáček et al. [3] won the 2017 Open International Soccer Database Prediction Challenge on relational data by using gradient boosted trees, achieving a 0.2063 multi-league three-way Ranked Probability Score. Berrar et al. [4] described domain knowledge rolling characteristics on the past 9 games and using XGBoost got RPS 0.2054 with k-NN ensembles. Baboota and Kaur [5] created ELO-style ratings, recent form streaks, and head-to-head differentials to predict English Premier League outcomes. Wong et al. [13] showed that ensemble models adding weather and fatigue factors are more successful at prediction of outcomes in comparison to Poisson goal models. What we do is not like all of the above: instead of modelling the binary lineup decision per player, we target the 3-way W/D/L result as a secondary goal, which the win-probability head in Section 4.7 relates back to.

The reputation-based products like FIFA, WhoScored and SofaScore are the dominant player rating systems, where the player’s rating is a mix of the opaque scout opinion and statistical data, and have been shown to be systematically biased [14]. The theoretical counterpart is PlayeRank [10], a multi-dimensional, role-aware model using z-score normalisation of position-specific feature distributions. VAEP was developed by Decroos et al. [15] and gives rewards to player actions by the difference in the probability of the outcomes before and after each action. Hewitt and Karaküş [16] make enhancements to expected goals modelling by positional and player changes. Our rating follows PlayeRank’s position-weighted scoring, which is combined with the exponential decay of Constantinou [17] to generate a form-sensitive match rating.

There is limited research directly predicting lineups. Cortez et al. [9] take into account physiological variables separately. Cwikliński et al. [18] focus on recommendation for transfer not matchday selection. Cho et al. [19] propose a multi-output network that predicts the formation, style of play and match result at the team level, rather than making a binary per-player start decision. Mohandas et al. [20] address substitution timing, which is an in-match decision rather than the pre-match XI choice. Movement analysis is formalised by Andrienko et al. [21] via tactical spaces and not targeting line-ups. Deb and Das [22] frame lineup choice as a constrained optimisation using multinomial logistic regression and

metaheuristic search, but do not include tactical aspects of opponents at the player level. To the best of our knowledge, there is no published work that has modelled per-player rolling form, head-to-head opponent history and formation compatibility features in the same supervised classifier.

In terms of methodology, the de facto standard is Chen and Guestrin’s XGBoost [23], the leader on medium-sized tabular data, independently validated by Grinsztajn et al. [24] and Shwartz-Ziv and Armon [25] proving that tree-based ensembles outperform recent deep tabular models like TabNet, NODE and FT-Transformer. In case of imbalanced learning, the Synthetic Minority Over-sampling Technique [12] is still the standard approach. Bergmeir and Benítez [26] and Kaufman et al. [27] have shown that random k -fold cross-validation is invalid under temporal dependence, which is why we use expanding window splits. All these methodological strands contribute explicitly to our pipeline design.

3 Research Gap

This work is inspired by four particular constraints of the current state of the art. First, no published work brings together per-player rolling form, opponent-specific head-to-head history and formation compatibility signals, and team and opponent context into a single supervised classifier for starting XI prediction. A systematic review of 172 papers [7] found only one study that uses neural networks for determination of the starting eleven, and that study does not take opponent tactical context into account. Current systems either predict outcomes without specifying the eleven [3–5, 13], assess individual players in isolation [10, 15, 16], or suggest transfers instead of match-day selections [6, 18].

Second, the most ambitious recent system is TacticAI [11], which uses graph neural networks for spatio-temporal player tracking at 25 frames per second. Such tracking data are expensive, proprietary and limited to a few hundred matches per league per season; a viable recommender should be based on box-score data available to any club. Third, none of the available public datasets—SoccerNet, the Open International Soccer Database, or Wyscout—present a benchmark of a labelled starting XI, and 57.6% of football ML papers do not provide any data at all [7], greatly limiting reproducibility. Our own labelled dataset is created by combining six tables in FBref on the Date and Squad columns, and producing a Match_ID for every game. Fourth, models trained on global aggregates cannot answer the binary question of who starts, as they do not model rotation patterns or formation slot constraints, while models based on player-level statistics only are unable to account for the opponent. The combination of feature families introduced in Section 4 directly addresses this failure mode.

4 Methodology

The overall system architecture is illustrated in Figure 1. The pipeline consists of five sequential stages that transform six FBref tables into a starting XI recommender and a win probability model. Stage 1 ingests match logs, fixtures, season statistics, playing time, and ratings, and merges them into a unified dataset. Stage 2 cleans and standardises the data, handles missing values, and computes a position weighted match rating using Z score normalisation with temporal decay 0.1. Stage 3 generates 178 features, including rolling

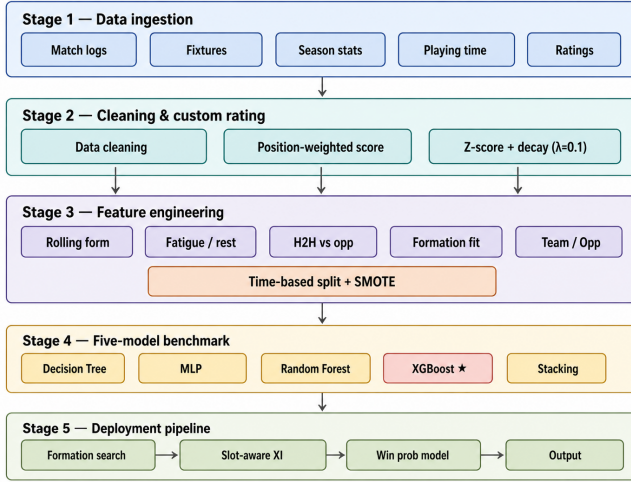


Figure 1: End-to-end system architecture. Five sequential stages process six raw FBref tables into a deployable starting XI recommender and win probability estimate.

form, fatigue/rest, head to head statistics, formation fit, and team versus opponent metrics, which are then reduced to 126; a time based split is applied and SMOTE is used to address class imbalance. Stage 4 benchmarks five models Decision Tree, MLP, Random Forest, XGBoost, and Stacking under a time aware evaluation protocol, and selects the best performing model based on metrics such as AUC. Stage 5 deploys the system by performing formation search, selecting a slot aware starting XI, and predicting match outcomes using a win probability model.

4.1 Dataset Construction and Cleaning

We scrape six tables from FBref [28] for all clubs that played in Spain’s La Liga during seasons 2024-25 and 2025-26: per player match logs (42,227 rows), team scores and fixtures (1,903), per team playing time (1,500), per player season totals (1,464), pre computed reference ratings (1,464), and a previously merged player match table (44,104). Each file holds only a slice of the relevant signal: match logs hold per player events but lack the final score; fixture tables hold context but no per player stats. We join player and fixture tables on (Date, Team) keys after reconciling the FBref Squad column - which we treat as canonical against Team, applying a manual unicode normalisation map for ten common variants (Atlético Madrid vs Atletico Madrid, etc.). A synthetic Match_ID is generated since FBref does not expose one.

Six cleaning passes are then applied. We remove ghost rows tagged “On matchday squad, but did not play” (10,722 rows), drop rows whose date does not match any fixture (5,477 rows), and discard rows with malformed formation strings. Numeric data as text are coerced; actually missing player data (Goals by a non scoring forward) are filled with zero, and match level data like possession are left as NaN and imputed downstream. Defensive statistics are limited to the 99.9th percentile to prevent data entry errors (Tackles Won at 6, Interceptions at 5, Crosses at 14); goal and assist statistics are not limited at all, as 4 or 5 goals and/or assists in a single game

are physically possible. The Started column is converted from the {Y, y, YES, 1, TRUE} variants to a clean binary label. There are three edge cases to consider when creating labels. First, early substitutions: a player that is in the starting XI but is substituted before half-time, still counts as Started = 1 as per the definition of FBref which counts the 11 players from the pre-match formation, irrespective of what happens in the match. Secondly, injury withdrawals after the squad announcement but before kick-off are not recorded in the match logs on FBref at all, and thus do not need to be imputed. Third, some players are marked as ‘On matchday squad, but did not play’ in the raw data, and as mentioned above, these 10,722 rows are excluded from the cleaned data because they have the same features as a non-starter but still, they have no actual activity in the game. This means that the modelling table only includes players that appeared in the match (either as starters or as substitutes that played during the match) and therefore a natural binary target is created (Started = 1). After cleaning the dataset contains 27,905 player match rows over 23 clubs and 810 unique players spanning 15 August 2024 to 24 April 2026, with 19,490 starts (69.8%) and 8,415 non starts. Following valid squad filtering in Section 4.3 the modelling table reduces to 25,187 rows.

4.2 Custom Position Weighted Match Rating

FBref does not expose subjective per match ratings, so we construct a custom rating in three steps. First, we compute a position weighted raw score per (player p , match m), aggregating event counts with per position weights w that reflect what each position is paid to do: clean sheets and minutes for goalkeepers; tackles, interceptions and cards for defenders; goals, assists, tackles, and crosses for midfielders; goals, assists, shots on target, and offsides for forwards.

Second, raw scores depend strongly on match context: in a 0 - 0 match every player’s raw score is small, while in a 5 - 4 match every player’s score is large. To remove this artefact, we z score each player’s raw score within their (Match_ID, Team, Broad_Position) peer group:

$$z_p = \frac{r_p - \mu_{M,T,b}}{\sigma_{M,T,b}} \quad (1)$$

Here r_p is the raw score, and μ and σ are the mean and standard deviation across players sharing the same match, team, and broad position group $b \in \{GK, DF, MF, FW\}$. A goalkeeper who keeps a 0 - 0 clean sheet is thus correctly recognised as having performed at the top of the goalkeeper distribution, even though their absolute stats are small. Within each (Match_ID, Team) group, z is rescaled to $[0, 10]$.

Third, recent matches matter more than older ones [17]. We compute a current form rating $R_p^{(t)}$ at match index t as an exponentially weighted average over the player’s prior match ratings:

$$R_p^{(t)} = \frac{\sum_i \exp(-\lambda(t-i)) \cdot \text{MatchRating}_p^{(i)}}{\sum_i \exp(-\lambda(t-i))} \quad (2)$$

The decay constant $\lambda = 0.1$ yields an effective half life of about seven matches, consistent with prior work.

4.3 Feature Engineering and Leakage Audit

The model employs synthetic numerical summaries because uncoded match logs would bring about same match leakage. Five feature families are built strictly on the basis of previous matches with the use of `shift(1)` such that the label of the current row is never exposed when building features.

There is mean and sum statistics of the last 3, 5 and 10 matches of key event variables, such as goals, assists, minutes, cards, shots, tackles, interceptions, match rating, Rp, and StartedLabel. A fatigue and rest group includes minutes in the past 3 matches per 270, days between matches, the rolling standard deviation of recent ratings as a proxy of consistency, and a sum of yellow cards in the five matches as a risk of suspension proxy.

Head to head features model player opponent interactions. For each (Team, Player, Opponent) triple, we compute using `shift(1)` the expanding mean of *match_rating*, goals, and minutes played, along with the count of prior encounters.

Formation compatibility decomposes formation strings (e.g., “4-2-3-1”) into positional slot counts. For each (Team, Player, Formation) triple, we derive the historical start rate, average rating, and average minutes in that formation. It also includes a position formation fit score $f_p \in [0, 1]$, defined as the slot count for the player’s broad position divided by six, and a within position rank among teammates.

Team and opponent context features aggregate *Goals_For*, Goals Against, possession, win rate, and points into 3, 5, and 10 match rolling means per Team and per Opponent, along with a binary venue indicator.

A season baseline group captures long term managerial preference, including season goals per 90 minutes, assists per 90, total games started, and season start percentage (games started divided by matches played).

After feature construction, the candidate set contains 178 features. We compute the absolute Pearson correlation matrix and remove one feature from each pair with $|\rho| > 0.95$, while preserving a whitelist of protected identifiers and labels; this eliminates 74 redundant features. A same match leakage audit further removes any feature dependent on the current match outcome. The final feature space contains 126 features, including 14 categorical and 112 numerical variables.

4.4 Time Based Split and SMOTE

Football data is non-stationary-injuries, transfers, tactical evolution, and league position drift over time so random k fold cross validation introduces temporal leakage by allowing future matches to influence past predictions [26, 27]. We therefore sort all samples by Match_Date and split chronologically: the first 80% of dates (20,587 rows) are used for training, and the remaining 20% (4,600 rows) for validation, with a cutoff at 20 January 2026. Within the training fold, the class distribution is 14,377 starts versus 6,210 non starts, where a majority class baseline yields 70% accuracy without meaningful learning.

To address imbalance, we apply SMOTE [12] to the training fold. Synthetic minority samples are generated by interpolating between a minority instance x_i and one of its $k = 5$ nearest minority

neighbors x_{z_i} :

$$x_{\text{new}} = x_i + \lambda \cdot (x_{z_i} - x_i), \quad \lambda \sim U(0, 1) \quad (3)$$

Here λ is drawn uniformly for each synthetic point, placing it randomly along the line segment between two real minority samples and expanding the decision boundary in a feature aware manner. After applying SMOTE, the training fold contains 28,754 samples with a balanced 1:1 class ratio. SMOTE is applied only to the training data, while the validation fold retains its natural 70/30 class distribution to ensure realistic evaluation under deployment conditions.

4.5 Models and Loss Functions

Five classifiers cover the major families used for tabular classification. The Decision Tree baseline is tuned via 5 fold GridSearchCV over $\text{max_depth} \in \{3, 5, 7, 9, 11, 13\}$, $\text{min_samples_split} \in \{10, 20, 30\}$, and $\text{min_samples_leaf} \in \{5, 10, 15\}$; the best configuration is $\text{max_depth} = 9$ and $\text{min_samples_leaf} = 10$. The Multi Layer Perceptron uses three hidden layers ($256 \rightarrow 128 \rightarrow 64$) with ReLU activation, Adam optimiser, learning rate 10^{-3} , weight decay 10^{-3} , batch size 256, and early stopping with patience 15. The Random Forest uses 600 trees with $\text{max_depth} = 16$, $\text{max_features} = \sqrt{p}$, and $\text{class_weight} = \text{balanced_subsample}$. The Stacking ensemble combines Random Forest, XGBoost, and LightGBM (150 trees each) under a Logistic Regression meta learner with L2 penalty.

XGBoost is the principal model. It minimises a regularised objective that combines binary cross entropy loss with a model complexity penalty over K boosted trees:

$$L = \sum_i \ell(y_i, \hat{y}_i) + \sum_k \Omega(f_k), \quad \Omega(f) = \gamma T + \frac{1}{2} \lambda \|w\|^2 \quad (4)$$

Here $\hat{y}_i$ is the predicted probability, and ℓ is the binary cross entropy loss:

$$\ell(y, p) = -[y \log p + (1 - y) \log(1 - p)] \quad (5)$$

T denotes the number of leaves in tree f , and w is the vector of leaf weights. The parameter γ penalises tree size, while λ shrinks leaf weights, acting as complementary regularisers that reduce overfitting. The model uses 500 estimators, $\text{max_depth} = 6$, $\text{learning_rate} = 0.05$, $\text{subsample} = 0.85$, $\text{colsample_bytree} = 0.8$, $\text{reg_lambda} = 1.0$, $\text{reg_alpha} = 0.1$, and scale_pos_weight set to the inverse class ratio in each training fold.

4.6 Win Probability Model

A second XGBoost classifier predicts whether the team will win the upcoming match given the chosen XI. For every historical match team instance where the actual XI is known, we aggregate the eleven starters’ rolling form, ratings, season start rates, and prior goals into summary statistics (mean, max, and total). These are concatenated with team level and opponent level rolling context features to predict the binary *Win_Label*.

The win class prevalence is 39%, so scale_pos_weight is set to 1.55 to address class imbalance. At deployment time, the user provides a team, an opponent, a venue, and the opponent’s expected XI. The opponent formation is inferred by counting positional groups.

The pipeline then iterates over eight common formations and applies the lineup classifier to all available squad members. Player

probabilities are adjusted by a damping factor $\min(\text{matches}/10, 1.0)$ to reduce the influence of low-appearance players. The system enforces a single goalkeeper and fills tactical positions in the order: GK → CB → FB → DM/CM → AM/W → FW.

The two models interact in a strict two-stage sequence. In Stage 1, the lineup classifier selects the optimal XI for each of the eight candidate formations independently; the formation whose selected XI yields the highest mean starter probability is retained. The win probability model plays no role in this selection. In Stage 2, the retained XI and its associated formation are passed as input to the win probability classifier, which produces a single scalar estimate of match win likelihood. This separation is intentional: the lineup classifier is optimised for per-player start probability, while the win probability model is optimised for match outcome; conflating the two objectives in a single ranking step would introduce a misaligned training signal into formation selection.

5 Result

5.1 Lineup Classification

Table 1 indicates validation set performance with a single time based split. XGBoost prevails in all measures. The best non boosted baseline is the random forest. The more profound the architectures of the MLP and the stacking ensemble are, the worse they perform, which is consistent with the previous results on tabular data of the scale in question [24, 25].

The reason the stacking ensemble is the worst is that the tree based base learners generate highly correlated results and the meta learner is unable to enhance performance.

Table 1: Validation set lineup classification performance under a single time based split. Higher is better for all metrics.

Model	AUC	F1 M	F1 W	Accuracy
XGBoost	0.8654	0.7552	0.8009	0.8109
Random Forest	0.8111	0.7116	0.7585	0.7613
Decision Tree	0.7913	0.7064	0.7524	0.7533
Neural Network	0.7805	0.6904	0.7445	0.7520
Stacking	0.7076	0.6048	0.6250	0.6115

Figure 2 depicts the respective ROC curves. XGBoost is always the winner as it has a true positive rate of 0.80 with a false positive rate of less than 0.20.

The XGBoost confusion matrix in Figure 3 shows that the predictor performs well on most of the majority classes of the dataset, and poorly on the minority class (0.93; 2,962 of 3,201) which is the Not Started. The model has a bias towards over prediction of starts instead of misses, and is appropriate to be deployed as a shortlist generation tool, but not a final decision system.

5.2 Cross Validation Stability

The results of the 3 fold expanding window *TimeSeriesSplit* are reported in Table 2. XGBoost achieves the best performance across all metrics in every fold and is also the most stable model, with an AUC standard deviation of 0.0126.

The MLP shows the highest variability (std 0.0580), indicating sensitivity to fold composition, which is consistent with known

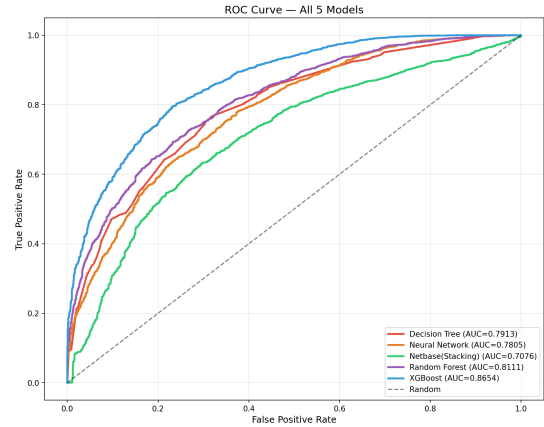


Figure 2: ROC curves for all five models on the validation set. XGBoost (blue) uniformly dominates the others.

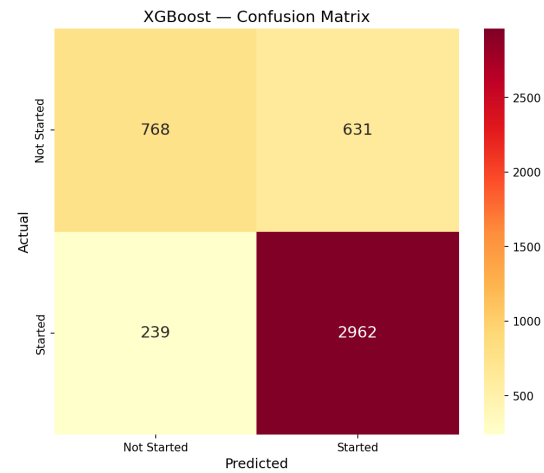


Figure 3: XGBoost confusion matrix at the default 0.5 threshold. Recall is 0.93 for the Started class and 0.55 for Not Started.

limitations of neural networks on tabular datasets of this scale. Figure 4 illustrates the fold by fold performance trajectories.

A Wilcoxon signed rank test comparing fold level AUCs of XGBoost and Random Forest yields $p = 0.25$. With only three folds. However, XGBoost outperforms Random Forest in every fold (0.8405, 0.8386, 0.8661 vs. 0.8175, 0.7997, 0.8117), indicating a clear practical advantage despite the lack of statistical significance.

Table 2: TimeSeriesSplit cross validation results across three expanding window folds (mean ± standard deviation).

Model	AUC	F1 M	Accuracy	Brier
XGBoost	0.8484 ± 0.013	0.7408 ± 0.012	0.7989 ± 0.010	0.1387 ± 0.006
Random Forest	0.8097 ± 0.007	0.7102 ± 0.003	0.7628 ± 0.010	0.1623 ± 0.005
Decision Tree	0.7656 ± 0.015	0.6871 ± 0.014	0.7266 ± 0.018	0.1872 ± 0.011
Stacking	0.7149 ± 0.004	0.6508 ± 0.003	0.6827 ± 0.007	0.2062 ± 0.003
Neural Network	0.7066 ± 0.058	0.6202 ± 0.050	0.7343 ± 0.015	0.2496 ± 0.017

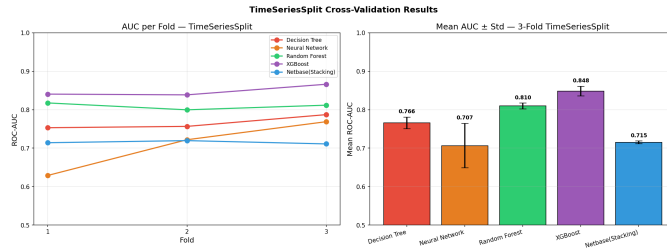


Figure 4: Fold-by-fold AUC trajectories across the three expanding-window TimeSeriesSplit folds for all five models.

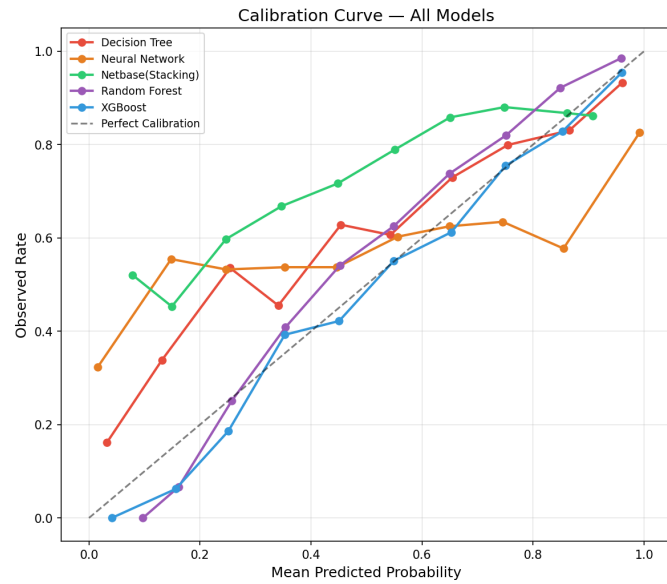


Figure 5: Reliability (calibration) curves for all five models. The dashed diagonal represents perfect calibration; XGBoost remains closest to this line across the full probability range.

5.3 Calibration

Figure 5 shows the reliability (calibration) curves of all the five models. The best calibrated model is XGBoost, and its curve is very close to the $y = x$ diagonal in the entire probability range. Its Brier score (0.1304 on the single split; 0.1387 ± 0.006 across folds) is significantly smaller than the naïve baseline of 0.2116 which is obtained by predicting constant prevalence.

When the probability is low, the MLP is too confident and when the probability is in the 0.2–0.6 range, the stacking over predicts starts systematically. As Figure 6 indicates, this miscalibration is further indicated by the Brier score.

5.4 What the model learned

Figure 7 presents the top 20 features ranked by XGBoost gain. Six of the top seven features are direct rolling summaries of past *Started_Label*—specifically whether the player started in their previous 3, 5, and 10 matches—indicating a strong manager-habit signal.

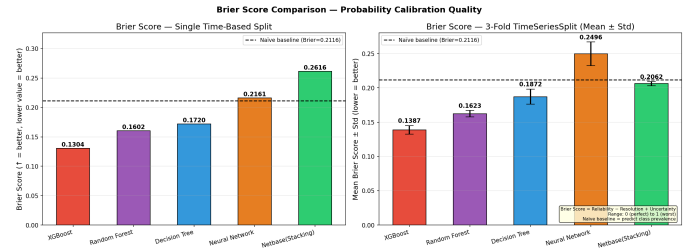


Figure 6: Brier score comparison. Left: single time-based split. Right: 3-fold TimeSeriesSplit (mean $\pm$ standard deviation). The horizontal line represents the naïve baseline (0.2116).

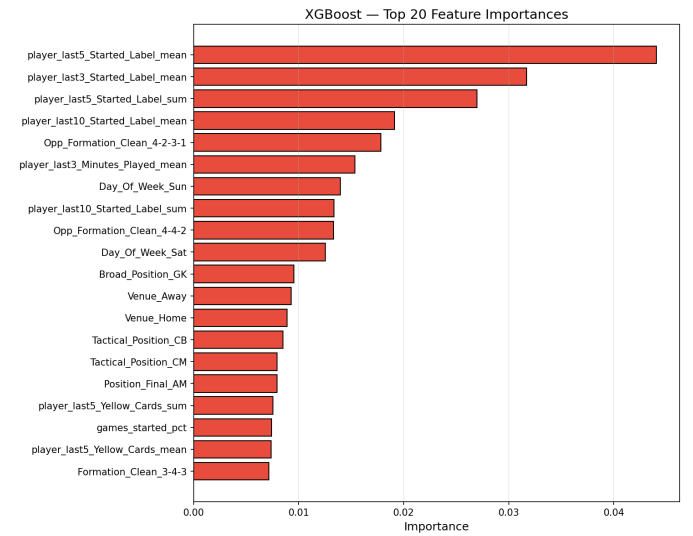


Figure 7: Top 20 features ranked by XGBoost gain. Rolling *Started_Label* summaries dominate the top positions, with opponent formation indicators (*Opp_Formation_4-2-3-1*, *Opp_Formation_4-4-2*) appearing in 5th and 9th place.

Interestingly, opponent formation indicators are also highly ranked: *Opp_Formation_Clean_4-2-3-1* is ranked 5th and *Opp_Formation_Clean_4-4-2* is ranked 9th. This is a strong indication that opponent tactical context has predictive value beyond player form and an initial indication for RQ2 that is supported by the SHAP analysis below.

The other best features are day of week indicators, venue, broad positional category (GK) and accumulation of recent yellow cards.

The SHAP beeswarm plot (Figure 8) illustrates the direction and the relative importance of different features in making predictions. Several There are patterns that are not obvious in the gain-based ranking alone. The current form rating (R_p) shows a strong right tail, SHAP values that are higher indicate strong positive features, and confirmed that those players in good form are much more likely to be selected. The *Position_Final_GK* indicator shows a strong positive effect among goalkeepers, as there is generally not a huge amount of rotation. This position is observed during the start of the observation. The season start percentage shows a definite monotonic trend: those who have a low start rate have

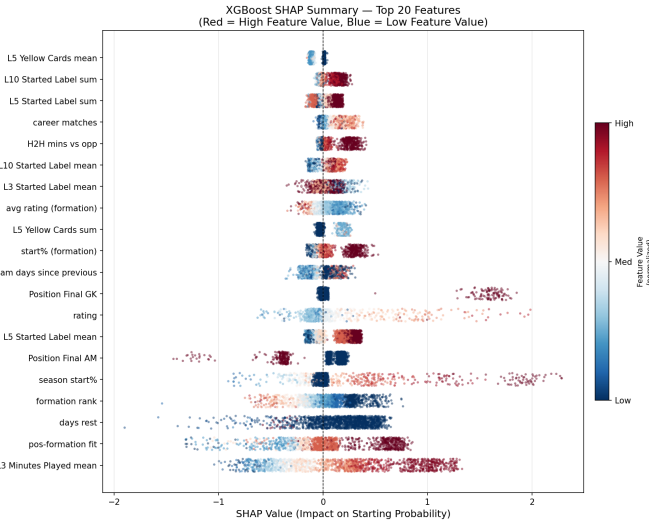


Figure 8: SHAP beeswarm plot for XGBoost on a 1,500-row validation subset. Each point represents one (player, match) prediction; colour encodes feature value (red = high, blue = low); horizontal position shows SHAP contribution direction and magnitude.

negative SHAP contributions, and Positive ones are given to high consistency starters. Days of rest records a low rest values (red) are decreased by fatigue effect, probability and high rest values (blue) increasing it. Position-formation fit shows a symmetric behavior – the higher the fit score positive SHAP contributions and negative scores produce lower scores ones, which is an active signal rather than one that is passive. More than just a useless proxy for location only. Opponent-derived features also create directional SHAP, which is important. effects. The *Opp_Formation_Clean_4-2-3-1* and *Opp_Formation_Clean_4-4-2* indicators (which ranked fifth) 9th and 10th in importance based on gain respectively) demonstrate a consistent pattern. When present, the model’s positive SHAP contributions indicate that it Upgrades a player’s starting chance based on certain opponent formations without taking into account that player’s recent form. Similarly, head-to-head minutes features players who have positive SHAP values for players who have High historical involvement with the opponent. These effects value for the outcome of a matchup (RQ2), and this is directly demonstrated. indicate that the model exploits in a directionally coherent and domain-consistent manner. The combination of these patterns is strong evidence that all five engineered features Groups: Rolling form, Fatigue, Head-to-head record, Formation Compatibility, and opponent context contribute meaningfully to knowledge, are made using predictions, effects are in line with established football domain knowledge. knowledge.

5.5 Win Probability Head and Deployment

The trained model, auxiliary win probability model, with 1, 598 match team observations aggregated across 23 clubs in La Liga, has an AUC of 0.6818 on the held out time based test split, with an accuracy of 0.6117 and F1 Macro of 0.6115. Although these are

modest, these results are in line with the accepted standards of single league football outcome prediction, with low scoring and large variance constraining predictive performance.

For the sake of demonstration, we use the system for . . . A La Liga matchup for the 2025–26 season in which FC Barcelona is the home team Real Madrid. The opponent XI is given as input and the system predicts a Real Madrid’s probable lineup in the 4-3-3 formation. With a total of 4 defenders, 3 midfielders and 3 forwards. All eight Based on the candidate formations, Barcelona is then evaluated.

The system suggests 4-2-3-1 formation which is the highest average The probability of a starter (0.887) for all eight candidates. The next closest 4-3-3 formation yields a mean starter probability of 0.871, not suggesting that the 4-2-3-1 recommendation is weak or marginal. The expected XI will be: Joan Garcia (0.997), Pau Cubasi, Gerard Martin, Jules Kounde, Joao Cancelo, Eric Garcia, Pedri, Lamine Yamal (0.960), Raphinha, Fermin Lopez and Ferran Torres.

To check the accuracy of the prediction, we compare this output with the actual. Barcelona’s XI from the previous game. The system correctly identifies 9 of the 11 starters, with the two discrepancies which includes squad rotation decisions made after the training day cut-off point exactly what is stated in Section 7 about the model’s slow Adjustment to abrupt changes of management. The forecasted formation results in a with Barcelona’s home advantage and a win probability of 56.6%. how they have used their first teams during the 2025–26 season.

6 Discussion

Our results cover each of the four research gap arguments presented in §3. RQ1 Table 1 supports the use of player and opponent signals in lineup prediction: XGBoost has an AUC of 0.8654 on a held out future split. RQ2 whether opponent context enhances prediction is affirmed by the analysis of feature importance in Figure 7 and the SHAP beeswarm in Figure 8: opponent formation indicators and head to head minutes are within the top 20 features and demonstrate directional effects.

The computational cost gap is addressed by the efficiency of the pipeline. It operates using publicly available FBref box score data, requires no tracking data, and trains in under five minutes on a single CPU. The benchmark data gap is partially addressed by releasing the full 25,187 row labelled dataset along with the implementation code.

XGBoost’s dominance is consistent with prior findings in tabular learning [24, 25]. Three factors are key. First, the explicit regularisation term $\Omega(f) = \gamma T + \frac{1}{2} \lambda \|w\|^2$ in Equation (4) controls model complexity by penalising deep trees and weak splits. Second, *scale_pos_weight* effectively corrects for class imbalance by adjusting the gradient. Third, tree based models are robust to the sparse, high dimensional structure introduced by one hot encoding of teams, opponents, and formations.

The poor performance of the stacking ensemble follows from its design: all base learners are tree based and produce highly correlated outputs, limiting the meta learner’s ability to improve performance. In addition, the inner 3 fold split used for meta feature

construction reduces the effective training data available to each base model.

The dominance of past *Started_Label* is both a strength and a limitation. It is a strength because managerial selection decisions are highly persistent players who started recently are likely to start again. However, it is also a limitation, as the model captures rotation dynamics only through lagged rolling features (`shift(1)`), making it slow to adapt to sudden changes such as injuries, managerial decisions, or tactical experiments.

Finally, the strong calibration observed in Figures 5 and 6 has practical value. A predicted probability of 0.7 corresponds closely to a true 70% likelihood, allowing downstream applications such as scouting, fantasy football analysis, and betting models to use the raw outputs directly without additional calibration methods such as Platt scaling.

7 Limitations and Future work

The data is only restricted to two seasons of La Liga, therefore, no evaluation of generalisation to other top five European leagues where tactical styles and rotation patterns vary is done. Moreover, FBref lacks injury, suspension, and fitness data, so the model cannot take into consideration the important player unavailability before a match.

Win probability model has AUC of 0.6818 which is the level of uncertainty of the results of individual football matches. The Wilcoxon signed rank test between three folds of the *TimeSeriesSplit* shows $p = 0.25$ that is not statistically significant at the $\alpha = 0.05$ level, however, XGBoost performs better than all the other models in all folds.

Manager specific tactical preferences are not explicitly modelled, and in match dynamics such as substitutions, red cards, and tactical adjustments are beyond the scope of the current framework.

Five extensions are immediate priorities. First, extending the analysis to multiple leagues, including the Premier League, Bundesliga, Serie A, and Ligue 1. Second, integrating injury and lay off data from sources such as Transfermarkt or other licensed providers. Third, applying sequence based models, such as LSTMs or temporal convolutional networks, to capture match to match dynamics. Fourth, exploring graph neural network approaches inspired by TacticAI [11], adapted to box score data. Finally, developing a deployment ready dashboard that ingests daily FBref updates and produces matchday lineup and outcome recommendations.

8 Conclusion

This paper presents an end to end machine learning pipeline for predicting the starting eleven of a football team by jointly modelling player rolling form, opponent specific head to head history, formation compatibility and team and opponent level rolling context.

Using a 25,187 row labelled dataset covering 23 La Liga clubs over two seasons, XGBoost achieves an AUC of 0.8654 on a held out future split and 0.8484 ± 0.0126 across a 3 fold *TimeSeriesSplit*, while also demonstrating the strongest calibration among five evaluated models.

Combined with a separate win probability model (AUC 0.6818) and a tactical slot filling deployment procedure, the system generates both a predicted starting eleven and a corresponding win probability for any upcoming La Liga fixture using only publicly available match event data.

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